PIN Generation Using String Manipulation

Write a Java program to generate a PIN based on two names, a six-digit PIN, and a position value (N).

The program should accept the first name, last name, a six-digit positive PIN, and an integer (N) as input. It must identify the longer and shorter names based on their lengths. If both names have the same length, the name that comes later in lexicographical order should be considered the longer name.

Construct a new string by concatenating:

1. The first character of the longer name.
2. The complete shorter name.
3. The (N)th digit of the PIN from the left.
4. The (N)th digit of the PIN from the right.

Finally, toggle the case of every alphabetic character in the constructed string: convert uppercase letters to lowercase and lowercase letters to uppercase. Digits must remain unchanged.

Input Validation Requirements:

- Both names must not be null or empty after trimming leading and trailing whitespace.
- The PIN must be a six-digit positive integer.
- The value of (N) must be between 1 and 6, inclusive.
- If any input is invalid, display an appropriate error message.

Sample Input:

```
First Name: Rajiv
Last Name: Roy
PIN: 560037
N: 5
```

Sample Output:

```
Generated PIN: rROy03
```

Explanation:

- The longer name is `Rajiv`, and the shorter name is `Roy`.
- The first character of the longer name is `R`.
- The 5th digit from the left in `560037` is `3`.
- The 5th digit from the right in `560037` is `0`.
- The constructed string is `RRoy30`.
- After toggling the case, the final generated PIN is `rROy30`.

Note: The correct sample output is `rROy30`, based on the specified rules.

Correct the inconsistent sample outputClarify the PIN input type